Name : Vivek Chaudhari

Project : 4

InfraForge : Script-based AWS infrastructure automation system

Project Overview (Simple)

InfraForge is a Python-based automation project that provisions AWS resources like EC2 instances and S3 buckets using scripts instead of manual console operations. It uses IAM for secure access and boto3 to interact with AWS services programmatically.

⚙️ AWS Services & Tools (Why Used + Script Explanation)

🔐 AWS Identity and Access Management (IAM)

Why used:
To securely control access to AWS resources.

Script / Explanation:

- Create IAM user or role
- Attach policies (like EC2FullAccess, S3FullAccess)
- Generate Access Key & Secret Key

👉 Needed because boto3 requires credentials

🖥️ Amazon EC2

Why used:
To create virtual servers using Python script.

Script / Explanation:

- Launch EC2 instance using boto3
- Define AMI ID, instance type, key pair
- Start/stop instances programmatically

👉 Example use: auto-deploy server

📦 Amazon S3

Why used:
To store files and manage storage via script.

Script / Explanation:

- Create S3 bucket
- Upload/download files
- Delete objects

👉 Example use: backup automation

🐍 boto3

Why used:
To interact with AWS services using Python.

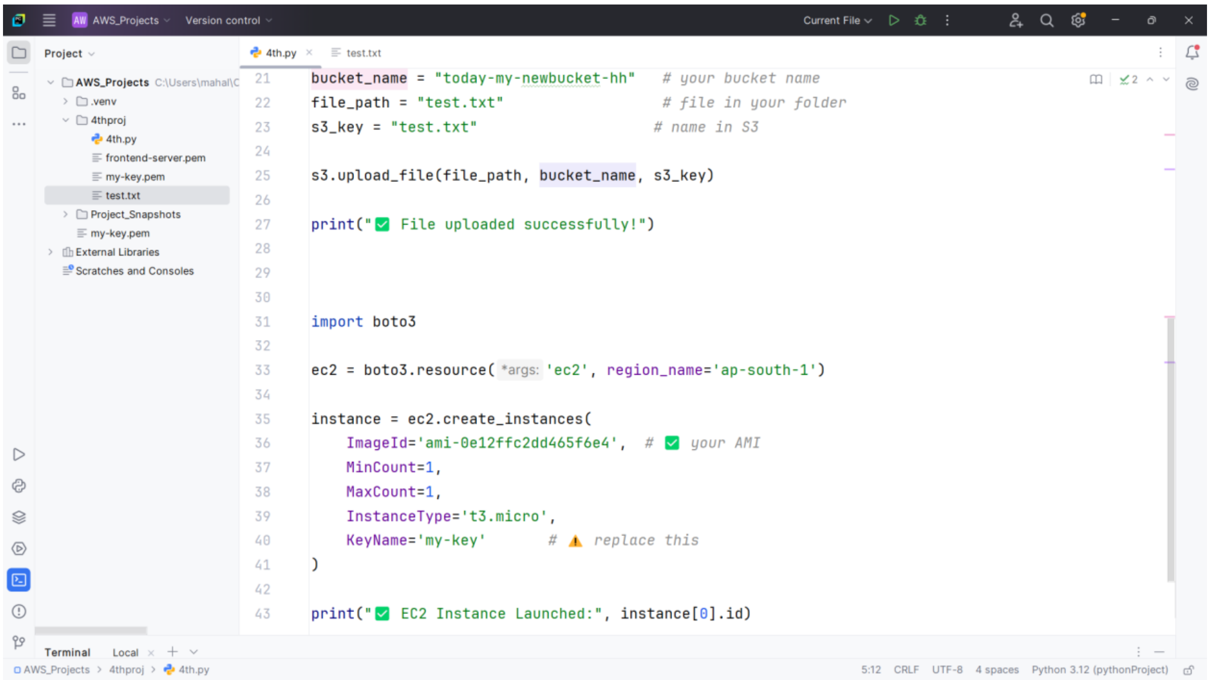
Script / Explanation:

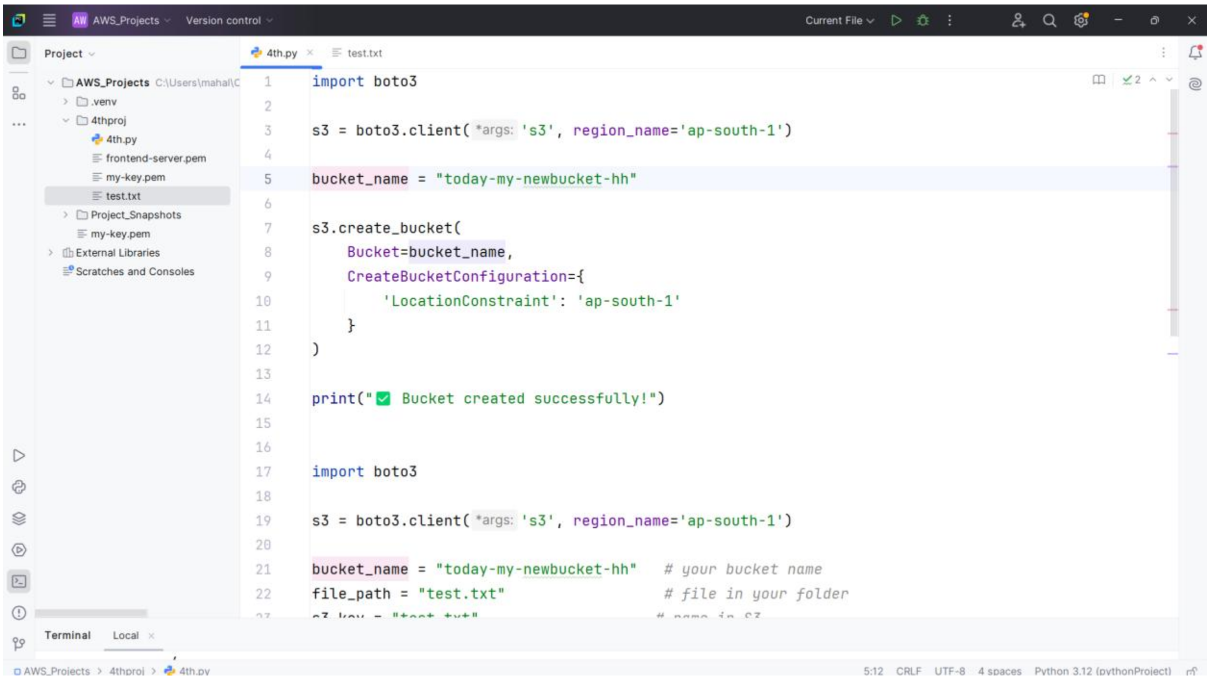
- Connect Python with AWS
- Call APIs like create_instance(), create_bucket()
- Automate full infrastructure

👉 Core of this project

🔄 Project Workflow

1. Configure IAM credentials
2. Write Python script using boto3
3. Script creates EC2 instance
4. Script creates S3 bucket
5. Script performs operations (upload/start/stop)


AWS_Projects
Version control
Current File
Project
4th.py
test.txt
AWS_Projects C:\Users\mahal\C
.venv
4thproj
4th.py
frontend-server.pem
my-key.pem
test.txt
Project_Snapshots
my-key.pem
External Libraries
Scratches and Consoles
bucket_name = "today-my-newbucket-hh"   # your bucket name
file_path = "test.txt"   # file in your folder
s3_key = "test.txt"   # name in S3
s3.upload_file(file_path, bucket_name, s3_key)
print("✅ File uploaded successfully!")
import boto3
ec2 = boto3.resource(*args: 'ec2', region_name='ap-south-1')
instance = ec2.create_instances(
    ImageId='ami-0e12ffc2dd465f6e4', # ✅ your AMI
    MinCount=1,
    MaxCount=1,
    InstanceType='t3.micro',
    KeyName='my-key'   # ⚠ replace this
)
print("✅ EC2 Instance Launched:", instance[0].id)
Terminal
Local
AWS_Projects > 4thproj > 4th.py
5:12 CRLF UTF-8 4 spaces Python 3.12 (pythonProject)


AWS_Projects
Version control
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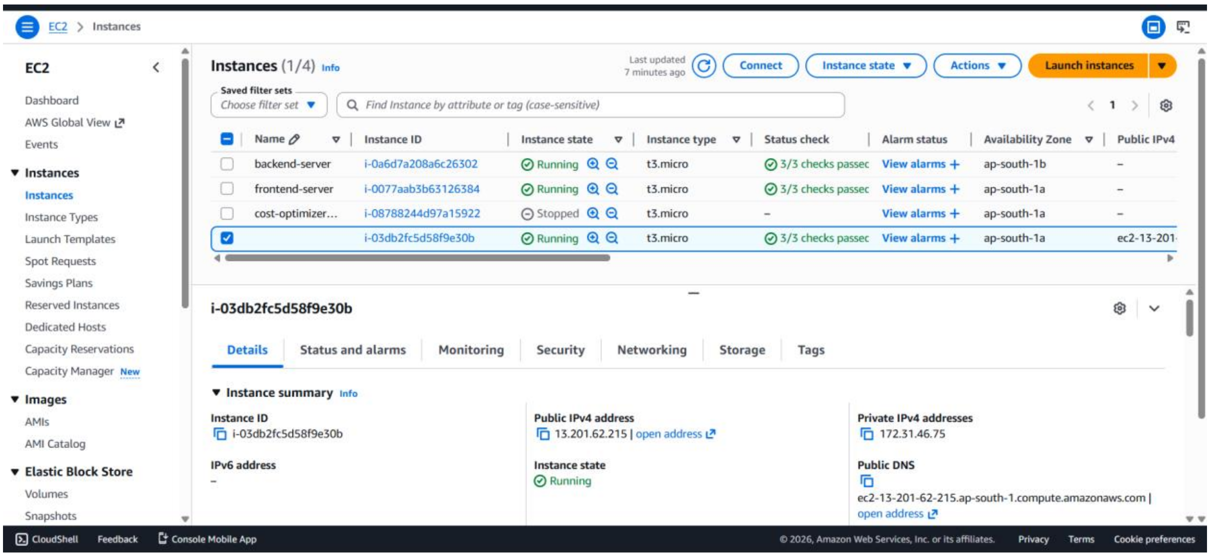
4thproj
4th.py
frontend-server.pem
my-key.pem
test.txt
Project_Snapshots
my-key.pem
External Libraries
Scratches and Consoles
import boto3
s3 = boto3.client(*args: 's3', region_name='ap-south-1')
bucket_name = "today-my-newbucket-hh"
s3.create_bucket(
    Bucket=bucket_name,
    CreateBucketConfiguration={
        'LocationConstraint': 'ap-south-1'
    }
)
print("✅ Bucket created successfully!")
import boto3
s3 = boto3.client(*args: 's3', region_name='ap-south-1')
bucket_name = "today-my-newbucket-hh"   # your bucket name
file_path = "test.txt"   # file in your folder
Terminal
Local
AWS_Projects > 4thproj > 4th.py
5:12 CRLF UTF-8 4 spaces Python 3.12 (pythonProject)


EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Instances (1/4) Info
Last updated 7 minutes ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Name | Instance ID | Instance state | Instance type | Status check | Alarm status | Availability Zone | Public IPv4
backend-server | i-0a6d7a208a6c26302 | Running | t3.micro | 3/3 checks passec | View alarms | ap-south-1b | –
frontend-server | i-0077aab3b63126384 | Running | t3.micro | 3/3 checks passec | View alarms | ap-south-1a | –
cost-optimizer… | i-08788244d97a15922 | Stopped | t3.micro | – | View alarms | ap-south-1a | –
i-03db2fc5d58f9e30b | Running | t3.micro | 3/3 checks passec | View alarms | ap-south-1a | ec2-13-201-
i-03db2fc5d58f9e30b
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-03db2fc5d58f9e30b
Public IPv4 address
13.201.62.215 | open address
Private IPv4 addresses
172.31.46.75
IPv6 address
–
Instance state
Running
Public DNS
ec2-13-201-62-215.ap-south-1.compute.amazonaws.com |
open address
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
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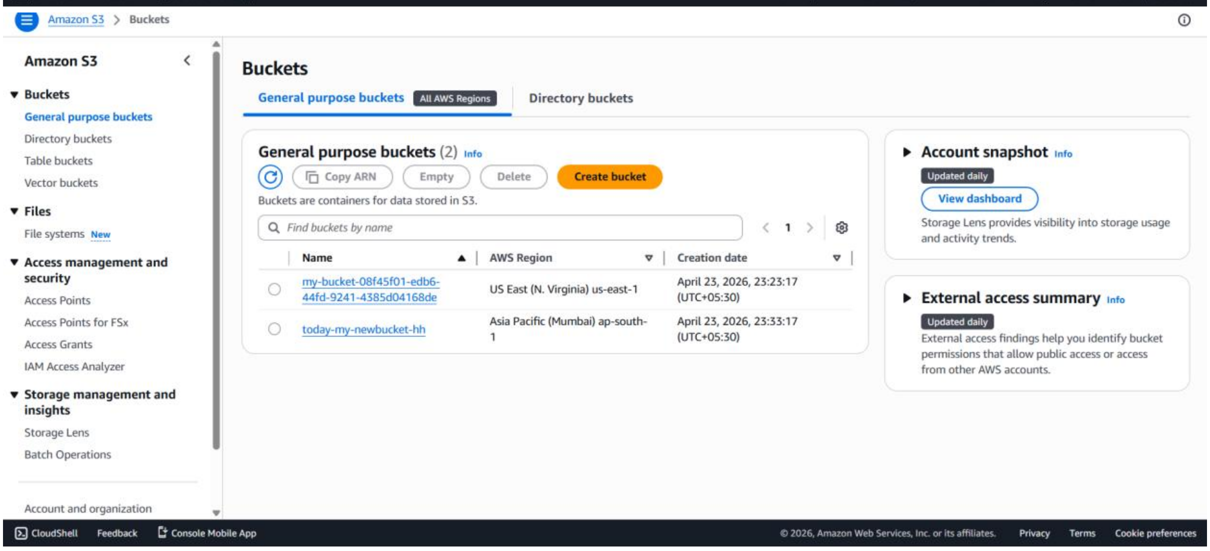
Terms
Cookie preferences


Amazon S3 > Buckets
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems New
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Account and organization
Buckets
General purpose buckets
All AWS Regions
Directory buckets
General purpose buckets (2) Info
Copy ARN
Empty
Delete
Create bucket
Buckets are containers for data stored in S3.
Find buckets by name
Name | AWS Region | Creation date
my-bucket-08f45f01-edb6-44fd-9241-4385d04168de | US East (N. Virginia) us-east-1 | April 23, 2026, 23:23:17 (UTC+05:30)
today-my-newbucket-hh | Asia Pacific (Mumbai) ap-south-1 | April 23, 2026, 23:33:17 (UTC+05:30)
Account snapshot Info
Updated daily
View dashboard
Storage Lens provides visibility into storage usage and activity trends.
External access summary Info
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
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